Ananth Jonnavittula

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Summary

Robotics researcher with 8+ years of experience across academic and industrial domains, specializing in perception-driven manipulation, visual imitation learning, and generative AI for robotics. Proven track record deploying production-grade vision systems and publishing impactful research in reinforcement and imitation learning. Seeking to develop on-foundational AI/ML technologies.

Technical Skills

Languages: Python, URScript, KUKA KRL
Frameworks/Tools: PyTorch, TensorFlow, OpenCV, ROS, Isaac Sim, Blender, TorchScript, GitHub
Robotics Platforms: Universal Robots, ABB, FANUC, KUKA, FrankaEmika, Fetch
Research Interests: Visual Imitation Learning, Robot Perception, Self-Supervised Learning, Human-Robot Interaction, Foundation Models

Education

Virginia Tech **Aug 2020 – Aug 2024**
PhD in Mechanical Engineering *Blacksburg, VA*

Worcester Polytechnic Institute **Aug 2015 – May 2017**
MS in Robotics Engineering *Worcester, MA*

SASTRA University **Jun 2011 – May 2015**
B.Tech in Electronics and Instrumentation *Tanjore, India*

Work Experience

Senior Research Engineer **Sep 2024 – Present**
Samsung Research America *Mountain View, CA*

- Developed VR based teleoperation systems for humanoid robots
- Designed universal teleoperation system stack for AI data collection
- Developed Nav2 based navigation for humanoid robots

AI Researcher **Dec 2024 – Sep 2025**
Foundation Future Industries Inc *San Francisco, CA*

- Developed Diffusion Policy based Imitation Learning policies for dexterous manipulation
- Created segmentation and pose estimation pipelines using DINOv2 and SAM2 for uncommon automotive parts
- Integrated NeRF-based SLAM pipelines for scene understanding in real world factories
- Engineered Nvidia Isaac Sim based simulation environment for custom humanoid robot

Graduate Research Assistant **Aug 2020 – Aug 2024**
Virginia Tech *Blacksburg, VA*

- Developed an imitation learning method that can learn long horizon tasks from a single video demonstration
- Implemented Soft Actor-Critic based Reinforcement Learning approaches for manipulation of YCB objects
- Engineered an inverse reinforcement learning based algorithm to learn from imperfect user demonstrations
- Spearheaded the development of a variational autoencoder based algorithm for imitation learning
- Conducted stability analysis for imitation learning methods in the context of shared autonomy

Research and Development Intern **Jun 2022 – Aug 2022**
ABB Inc *San Jose, CA*

- Implemented and analyzed methods that use stereo vision or RGBD images for instance segmentation
- Architected an instance segmentation algorithm for small parcels singulation using detectron2
- Devised a BlenderProc based pipeline to generate synthetic datasets for training models
- Engineered a pipeline to export trained models from Python to C++ using Torchscript

Robotics/Vision Engineer **Jun 2017 – May 2020**
Parker Hannifin Corp. *Haverhill, MA*

- Engineered an automated cell for palletizing over 100 different SKUs to reduce labor costs
- Designed and implemented an automated laser marker that doubled throughput in multiple manufacturing cells
- Developed a robotic cell using UR5 robot and cameras for part recognition and end capped filter
- Engineered and deployed a urethane end capping cell using FANUC robots

Engineering Intern **Jan 2017 – Apr 2017**
Parker Hannifin Corp. *Haverhill, MA*

- Conducted feasibility analysis on automated end capping using collaborative robots
- Programmed controllers for automated part feeding using vibratory feeders
- Implemented image recognition using Keyence CV-X series to detect orientation of parts

Graduate Research Assistant **Jan 2016 – May 2016**
Worcester Polytechnic Institute *Worcester, MA*

- Conducted analysis for range of motion and kinematic requirements for a 2 DOF hydro-muscle actuated leg
- Designed a coupling mechanism that locks the leg while maintaining pose in case of serious failure
- Orchestrated the development closed-loop control system for leg actuation using four pairs of hydro-muscles
- Developed controllers for hydraulics and coupling mechanism using Arduino microcontrollers

Publications

VIEW: Visual Imitation Learning with Waypoints **2025**
Springer Autonomous Robots
__Ananth Jonnavittula__, Sagar Parekh, and Dylan P. Losey

SARI: Shared Autonomy across Repeated Interaction **2024**
ACM Transactions on Human-Robot Interaction
__Ananth Jonnavittula__, Shaunak A. Mehta, and Dylan P. Losey

Communicating Robot Conventions through Shared Autonomy **2022**
International Conference on Robotics and Automation (ICRA)
__Ananth Jonnavittula__ and Dylan P. Losey

Here's What I've Learned: Asking Questions that Reveal Reward Learning **2022**
ACM Transactions on Human-Robot Interaction
Soheil Habibian, __Ananth Jonnavittula__, and Dylan P. Losey

Learning to Share Autonomy Across Repeated Interaction **2021**
IEEE International Conference on Intelligent Robots and Systems (IROS) *Best Student Paper Finalist*
__Ananth Jonnavittula__ and Dylan P. Losey

I know what you meant: Learning human objectives by (under)estimating their choice set **2021**
IEEE International Conference on Robotics and Automation (ICRA)
__Ananth Jonnavittula__ and Dylan P. Losey

Patents

Biologically inspired joints and systems and methods of use thereof **2020**
US010632626B2